

Smart, Portable AUTOMOTIVE DOPPLER RADAR

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Measuring the speed of moving vehicles and objects is critical for ensuring road safety, conducting

traffic analysis, and supporting research applications. Conventional systems are often bulky or costly, whereas current requirements

emphasise compact, efficient, and affordable alternatives.

This system offers a portable, wireless speed detection solution based on the Doppler effect. It analyses frequency shifts in millimetre wave signals caused by the relative motion between a moving object and the sensor. The system accurately measures the velocity of fast-moving targets such as cars, motorbikes, and trains through precise millimetre wave transmission and reception.

This smart Doppler radar system is designed for versatility and supports effective speed monitoring, whether mounted on a roadside pole or used as a portable unit



Fig. 1: Author's prototype being tested

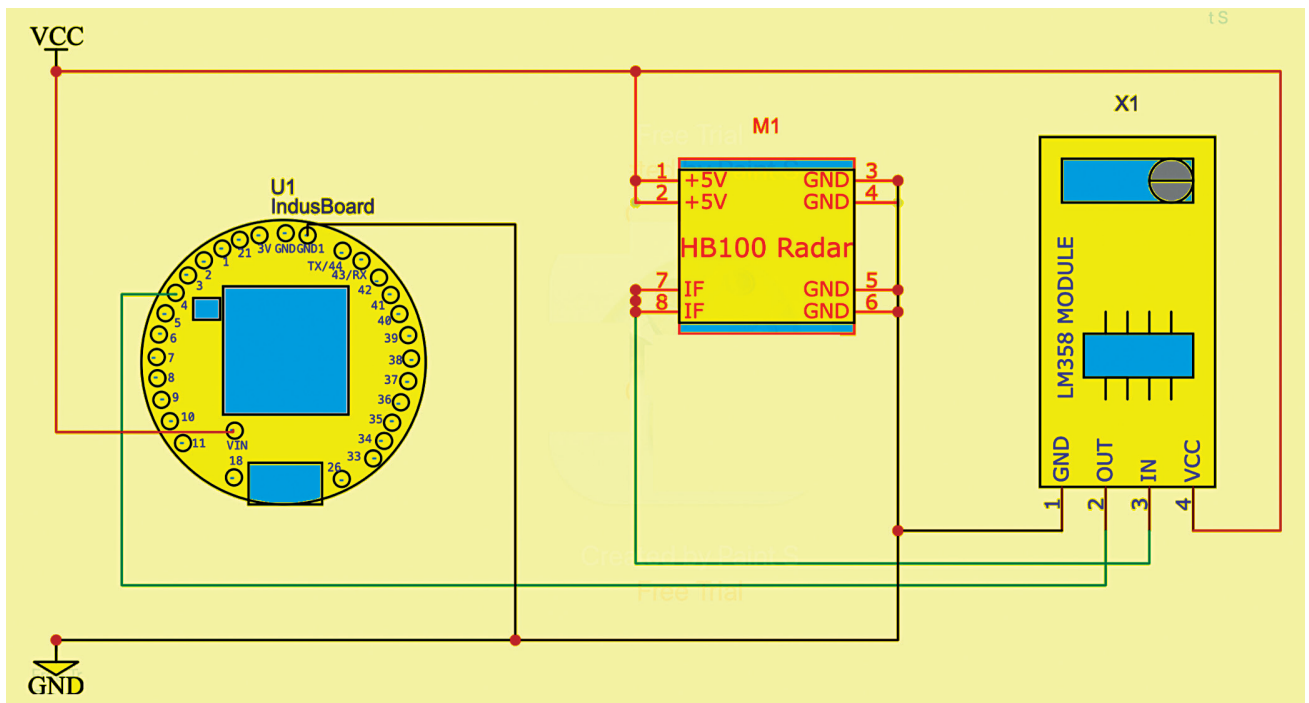


Fig. 2: Automatic Doppler radar circuit